

AGARD FLIGHT TEST MANUAL

VOLUME IV

PART IV C

POWER SPECTRAL METHODS OF ANALYSIS AND THEIR
APPLICATION TO PROBLEMS IN AIRPLANE DYNAMICS

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IVC:1 INTRODUCTION

The field of dynamics is largely concerned with the response of systems (mechanical, electrical, etc.) to various types of disturbances. A number of techniques exist for the analysis of certain types of disturbances and these have found wide application in many fields. The use of responses to step inputs, pulse inputs, and sinusoidal inputs is already familiar and useful in many aeronautical areas.

Fourier methods in particular have found wide regions of application in aeronautics and have provided many useful results. In these cases, the methods of analysis permit the synthesis of responses for linear systems to certain types of arbitrary but specified disturbances. However, if the disturbances are aperiodic and nondissipative, these methods will generally no longer apply. Furthermore, if the disturbances contain an element of chance or randomness and cannot be defined precisely, but only by certain probability distributions and average values, the foregoing methods can no longer be used and more general techniques are required.

The introduction of a chance or random element leads naturally to the application of probability and statistical notions which in combination with classical Fourier techniques have in recent years evolved into a general technique for the analysis of problems of this type. These techniques stem from the field of random process theory and from generalized harmonic analysis. Of particular appropriateness for present purposes are the methods centered around the use of power spectra.

The motivation and early development of power spectral techniques had its roots in studies of economic time series at the turn of the century. In the physical sciences, the first applications were concerned with the problems of light, the Brownian motion of particles, and turbulence. The development of a rigorous mathematical theory is even more recent and appears to be largely

due to Kolmogoroff and Wiener. Prior to and during the last war, many useful applications of these techniques were made in the fields of communications and servomechanisms.

The applications of these techniques to aeronautical problems appear to have been initiated since the last war and are thus still in an early stage of development. Already, the list of aeronautical problems to which these techniques have been usefully applied include:

- (a) Gust loads on airplanes
- (b) Buffeting loads
- (c) Acoustic noises
- (d) Tracking errors and human response studies
- (e) Wind tunnel turbulence
- (f) Runway roughness (land and sea)
- (g) Electronic noises in guidance and gunnery systems.

Refs. 1 to 24 provide a partial list of unclassified papers in this area. Although this list is already impressive, there will no doubt be a further rapid development in these areas and extensions to new areas of application.

In spite of the large amount of activity in these areas of aeronautics, there still appears to be some lack of appreciation of the basic motivation of power spectral techniques, the fundamental relations and their limitations, and some of the problems encountered in applications. This condition has in part arisen from the specialized nature of the applications so far made, the scattered nature of the literature, and the lack of definitive treatments in the engineering literature.

The application of these techniques in flight test studies also places some special

requirements on instrument characteristics, flight test techniques, and data reduction procedures. For these reasons, it appeared appropriate to include a paper in this volume which would at least survey the basic notions, the fundamental relations of these newer techniques and their relation to classical techniques, and some of the problems encountered in applications.

If the disturbance is characterized by certain probability distributions or average values, then it is necessary to consider all disturbances having these characteristics. Another way to say this is that we are concerned not with a single time function, but an ensemble of time functions having the required properties. We shall require an analytic representation for such ensembles of functions and rules for performing input-output or response calculations.

Also, in practice, finite and incomplete data are the rule; a small piece of one time history is frequently available and must be used to estimate results for more general situations. This is the familiar inference problem of the statistician. This matter, in the present case, turns out to be far from a trivial one and the problems that stem from it can be largely blamed for apparent inconsistencies in applications, and for the long delay in the practical applications of these general techniques. In fact, a clear appreciation of these problems and useful results in this respect are only very recent achievements (Ref. 25) and will be considered herein.

The order of presentation will start with a brief discussion of the classical techniques for response determination to arbitrary disturbances. The concepts of power spectra and the relations between the spectra and time-history characteristics will then be briefly reviewed and the relations between the power spectra of a disturbance and the response to the disturbance for a linear system will then be presented. The practical problem of measuring power spectra from finite data will then be considered and digital and analog techniques discussed. Finally in the interest of concreteness, some examples of results

obtained in aeronautical applications will be described.

IVC:2 RESPONSE DETERMINATION FOR ARBITRARY DISTURBANCES

In the way of background, some of the more important classical techniques that are used in the determination of the responses to arbitrary disturbances are briefly reviewed. Fourier techniques are described, as these techniques, when fused with statistical theory, form the basic mathematical structure for spectral methods. No effort is made to be complete, but rather the main ideas are given in a manner which permits a natural extension into the random process applications.

(a) Fourier Series

Under certain conditions, an arbitrary function $x(t)$ defined in the interval $T/2$ to $-T/2$ may be represented by a Fourier series expansion as follows:

$$x(t) = \sum_{n=1}^{\infty} a_n \sin \omega_n t + \sum_{n=0}^{\infty} b_n \cos \omega_n t \quad (1)$$

where

$$\omega_n = n\omega_0 \quad n = 1, 2, 3 \dots$$

$$\omega_0 = \frac{2\pi}{T}$$

$$a_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) \sin \omega_n t \, dt$$

$$b_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) \cos \omega_n t \, dt$$

$$b_0 = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) \, dt$$

The function defined by Eq. (1) is recurring with period equal to T . Thus

$$x(t) = x(t \pm kT) \quad \text{for } k = 0, 1, 2 \dots \quad (2)$$

It will be helpful to rewrite Eq. (1) into other equivalent forms; the complex form,

$$x(t) = \sum_{-\infty}^{\infty} C_n e^{i\omega_n t} \quad (3)$$

where

$$C_n = \frac{b_n - ia_n}{2}$$

$$C_0 = b_0$$

$$C_{-n} = \frac{b_n + ia_n}{2}$$

and in terms of amplitudes and phases

$$x(t) = \sum_{n=0}^{\infty} c_n \cos(\omega_n t - \psi_n) \quad (4)$$

where

$$c_n = \sqrt{a_n^2 + b_n^2}$$

$$\psi_n = \arctan a_n / b_n.$$

The representation of the arbitrary function $x(t)$ as the sum of a series of elemental sine and cosine terms permits the ready determination of the response $y(t)$ of a linear system (but only because of its linearity) to this disturbance by superposition, as follows:

$$y(t) = \sum_{-\infty}^{\infty} T(\omega_n) C_n e^{i\omega_n t} \quad (5)$$

where $T(\omega_n)$ is the frequency response

function for the system response to a specified disturbance and describes the (amplification and phase shift) response to the disturbance $e^{i\omega_n t}$. Eq. (5) may also be expressed in real form analogous to Eq. (4), as

$$y(t) = \sum_{n=0}^{\infty} |T(\omega_n)| c_n \cos(\omega_n t - \psi_n + \theta_n) \quad (6)$$

where the bars designate the amplitude of the complex quantity (amplification factor) and θ_n is the corresponding phase shift, so that

$$T(\omega_n) = |T(\omega_n)| e^{i\theta_n}.$$

(b) Fourier Integral

In order to extend the preceding results to nonperiodic phenomena, the limit of Eq. (3) is taken as $T \rightarrow \infty$, $\omega_0 \rightarrow 0$. The continuous form of Eq. (3) becomes

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega \quad (7)$$

where, as the limit is approached

$$C_n \rightarrow \frac{X(\omega)}{2\pi} d\omega$$

and

$$X(\omega) = \int_{-\infty}^{\infty} x(\tau) e^{-i\omega\tau} d\tau. \quad (8)$$

The quantity $X(\omega)$ is called the spectrum of $x(t)$ and $X(\omega) d\omega$ denotes the contribution of $y(t)$ to the frequency components between ω and $\omega + d\omega$. These relations are frequently written as

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{i\omega t} \int_{-\infty}^{\infty} x(\tau) e^{-i\omega\tau} d\tau \quad (9)$$

which is the well-known Fourier integral theorem.

For this representation to apply, it is of course necessary for the defining integral to be convergent. The essential condition is that as $t \rightarrow \infty$, $x(t) \rightarrow 0$ at a reasonably fast rate. This condition acts as a serious limitation on the use of this representation.

If similar limiting processes are carried out with respect to Eq. (5) as were applied to Eq. (3), the following result is obtained for the response

$$y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} T(\omega) X(\omega) e^{i\omega t} d\omega. \quad (10)$$

Eq. (10) gives the response of a system to the arbitrary disturbance $x(t)$. It is, of course, necessary for the defining integral to be convergent which in turn also requires that the disturbance $x(t)$ be dissipating and that the system be damped.

(c) Step and Impulse Response Functions

Another useful technique for the development of the response to arbitrary disturbances is based on the superposition of properly spaced responses to unit step and impulse disturbances. In these terms, the response to a disturbance starting at time zero is given by the well-known Duhamel's integral

$$y(t) = x(0)A(t) + \int_0^t \frac{dx(\tau)}{d\tau} A(t-\tau) d\tau \quad (11)$$

where $A(t)$ is the response to a unit step at time 0.

Integrating Eq. (11) by parts yields the equivalent form

$$y(t) = x(t)A(0) + \int_0^t x(\tau) A'(t-\tau) d\tau \quad (12)$$

where the function $A'(t)$ is the derivative of $A(t)$ with respect to time and is known as the unit impulse response.

If $x(0) = 0$ and $A(0) = 0$, Eqs. (11) and (12) may be written simply as

$$\begin{aligned} y(t) &= \int_0^t x'(\tau) A(t-\tau) d\tau \\ &= \int_0^t x(\tau) A'(t-\tau) d\tau. \end{aligned} \quad (13)$$

These relations are the input-output relations for an arbitrary but specified disturbance. The unit step response $A(t)$ and the unit impulse response $A'(t)$ are also simply related to the frequency response function, as follows:

$$\begin{aligned} T(\omega) &= \int_0^{\infty} A'(t) e^{i\omega t} dt \\ &= i\omega \int_0^{\infty} A(t) e^{i\omega t} dt \end{aligned} \quad (14)$$

provided that $\int_0^{\infty} |A'(t)| dt$ is finite.

In principle, Eq. (13) permits the determination of responses to arbitrary disturbances in a relatively straightforward manner, but this is usually complicated. Analytical solutions will only be practical in the simplest cases. When $x(t)$ has to be found, approximation of the integral equation by a system of many linear equations is not difficult, but solution of the linear equations requires the inversion of a matrix, which appears practical in only the nearly simplest cases.

IVC:3 RANDOM PROCESSES

So far, we have considered arbitrary but determinate (nonrandom) functions and have been able for certain cases to obtain their

representation by superposition of elemental type functions such as sines and cosines, unit steps, and impulses. There exist many disturbances in the physical, engineering, and social sciences as well as in aeronautics where the disturbance of interest is not so well defined. Fig. IVC:1 illustrates three examples of this kind that occur in aeronautics.

This indefiniteness arises from an element of chance or randomness which affects the details of the function. A simple example is furnished by the fluctuations of buffeting pressure as measured at a point on a wing. If this test were repeated, the exact time history would not be repeated exactly, but the essential character, averages, or other statistical characteristics could be expected to be unchanged.

If we are interested in the buffeting fluctuations, then our method of representation may not be very adequate if we describe only

one test result. We would instead prefer to use a representation which adequately describes the ensemble of functions that would be obtained if the test were repeated many, many times. This ensemble of functions is referred to as the random process. In many physical cases, this ensemble can be represented in a simple and useful manner by means of random process theory. Accounts of the mathematical theory of random processes may be found in Refs. 26 to 29.

In the present paper, we shall restrict ourselves to stationary processes; that is, functions whose characteristics do not change with time. This restriction is not too serious although useful results have already been obtained for nonstationary or evolutionary processes. However, the greatest simplification and usefulness is associated with the stationary processes.

Some of the results to follow are restricted to Gaussian processes. However, many

EXAMPLES OF RANDOM DISTURBANCES

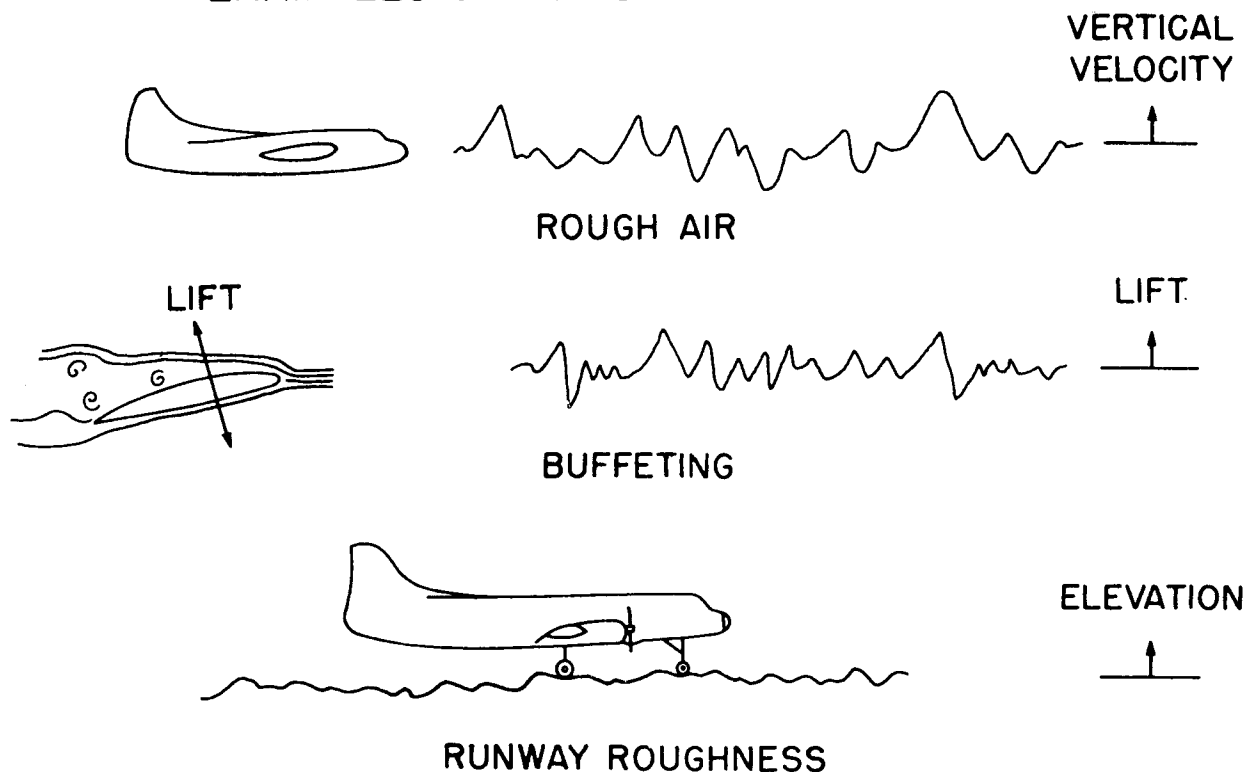


Fig. IVC:1 Examples of Random Disturbances

processes of aeronautical interest are closely approximated by Gaussian processes, and these results can usually be applied to good effects.

In order to provide a further insight into the concept of a random process, an example of a random process is given. The example chosen is a special one, involving the general form of the Fourier series expansion of an arbitrary function, and leads conveniently to the important case of a Gaussian random process.

(a) The Generation of Random Processes

If we make use of the Fourier series representation of Eq. (4), namely,

$$x(t) = \sum_{n=0}^{\infty} c_n \cos(\omega_n t - \psi_n) \quad (15)$$

where $\omega_n = n\omega_0$, it is possible, by imposing random characteristics to the coefficients c_n and ψ_n , to illustrate quite simply a means of generating a random process.

Thus, if we consider the single term

$$x(t) = c_n \cos \omega_n t \quad (16)$$

and suppose that $c_n = 0$ or 1 each with probability of 1/2, successive random choices of c_n lead to a series of functions, one-half of which are everywhere zero, the other half, unit cosines.

This example appears trivial but might apply to a simple on-off device. It should be noted that this process is not a stationary process since the average values at a given time depend upon time in a definite manner.

Consider next the function

$$x(t) = \cos(\omega_n t - \psi_n) \quad (17)$$

and permit the phase angle ψ_n to be random with uniform probability for the interval 0 to 2π .

We thus generate an ensemble of unit cosine functions. At any time t , the probability distribution of x (the relative frequency of various values of x over the ensemble) is given by

$$P(x) = \frac{1}{\pi} \frac{1}{\sqrt{1-x^2}} \quad -1 \leq x \leq 1 \quad (18)$$

Other statistical characteristics of this process may be likewise derived from Eq. (17). For any time t , we obtain, for example

$$\text{ave} \{x\} = 0$$

$$\text{ave} \{x^2\} = \frac{1}{2}$$

and we have

$$\text{ave} \{x(t_1) \cdot x(t_2)\} = \cos \omega_n (t_2 - t_1)$$

where the brackets designate that the average is taken over the ensemble.

This process is stationary inasmuch as its statistical characteristics do not depend upon the starting time.

(b) Approximately Gaussian Process

So far we have considered somewhat trivial processes. A more complicated stationary process may be obtained if use is made of Fourier series representation of Eq. (15)

$$x(t) = \sum_{n=1}^{\infty} c_n \cos(\omega_n t - \psi_n)$$

where without loss of generality we have assumed that $\bar{x} = 0$.

If it is assumed that the c_n values are fixed constants and the phase shifts ψ_n ,

are independent and, as before, random with a uniform probability distribution over the interval 0 to 2π , then an approximation to the interesting and important Gaussian random process is obtained. Inasmuch as $x(t)$ for each given time is the sum of a large number of random and independent variables with finite variance, it follows from the central limit theorem that for this time t , $x(t)$ approximates a Gaussian (or normal) distribution defined by

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/2\sigma^2} \quad (19)$$

where $\sigma^2 = \overline{x^2} = \Sigma c_n^2/2$ (the mean value of x^2).

The approximation to this distribution becomes arbitrarily good when the largest c_n becomes arbitrarily small. The distribution of Eq. (19) depends only upon the mean square value of x . It may also be shown that the distribution of x for various times t_1, t_2 , etc. (relative frequency over the ensemble of given combinations of values at t_1, t_2 , etc.) has a multivariate normal distribution.

For the case of two values of time, t_1 and t_2 , the joint probability distribution of x_1 and x_2 is given by

$$P(x_1, x_2) = \left(\frac{\sigma_1^2 \sigma_2^2 - R_{12}^2}{2\pi} \right)^{-\frac{1}{2}} \exp \left[\frac{-\sigma_2^2 x_1^2 - \sigma_1^2 x_2^2 + 2R_{12} x_1 x_2}{2(\sigma_1^2 \sigma_2^2 - R_{12}^2)} \right] \quad (20)$$

where

$$\sigma_1 = \sigma_2 = \sigma$$

and where

$$R_{12} = \text{ave} \{x_1 x_2\}$$

and (since the average values of x_1 and x_2 vanish) is termed the covariance of x_1 and x_2 .

The generalization to the multivariate distribution for t_1, t_2 , etc., is straightforward and can be shown to depend only on such covariances. Inasmuch as the process is a stationary one, the covariances $\text{ave} \{x_i x_j\}$ will depend only on the time difference $j-i$. Thus setting $j-i = \tau$, we see that the multivariate distribution depends only on the function $R(\tau)$, the autocovariance function.

In this sense, the autocovariance function $R(\tau)$ is said to define completely the Gaussian process. Also since the process is stationary, averages over time may, in practice, be taken as equivalent to ensemble averages and we may write

$$R(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) x(t+\tau) dt. \quad (21)$$

(In the engineering literature $R(\tau)$ is frequently called the autocorrelation function although this term more properly should be confined to $R(\tau)/R(0)$.)

The autocovariance function has the following properties:

- (a) $R(\tau) = R(-\tau)$
 - (b) $R(0) \geq R(\tau)$
 - (c) $R(0) = \sigma^2$
- (22)

Another function, which is even more useful for many purposes may be obtained by taking the Fourier transform of $R(\tau)$ as follows:

$$\Phi(\omega) = \frac{2}{\pi} \int_0^{\infty} R(\tau) \cos \omega \tau d\tau \quad (23)$$

where $\Phi(\omega)$ is the so-called energy or power spectrum of x and plays the central role in subsequent considerations.

The reciprocal relation also holds, namely

$$R(\tau) = \int_0^{\infty} \Phi(\omega) \cos \omega \tau d\omega \quad (24)$$

If Eq. (23) is rewritten in complex form as

$$\Phi(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} R(\tau) e^{i\omega\tau} d\tau \quad (25)$$

and Eq. (21) for $R(\tau)$, substituted in Eq. (25), another expression for the power spectrum may be derived which indicates the relation in the form of Fourier transforms. The formal result obtained is

$$\Phi(\omega) = \lim_{T \rightarrow \infty} \frac{1}{\pi T} X_T(\omega) X_T^*(-\omega) \quad (26)$$

where

$$X_T(\omega) = \int_{-T/2}^{T/2} x(t) e^{-i\omega t} dt \quad (27)$$

(Since $X_T^*(-\omega)$ is the complex conjugate of $X_T(\omega)$, the value of Eq. (26) is real and positive.)

The quantity $\Phi(\omega)d\omega$ can be considered to define the contribution to $R(0)$ (or σ^2) of the frequency components of $x(t)$ between ω and $\omega + d\omega$. Inasmuch as the power spectral density function is directly related to the autocovariance function, it likewise, and in the same sense, specifies the Gaussian random process. We shall later see how in the Gaussian case the spectrum is simply related to some of the more important statistical characteristics of the time history. For the non-Gaussian case, it also describes the distribution of the power with frequency

which is, in general, an important property of a random disturbance.

The power spectral density function has the following properties:

$$(a) \quad \Phi(\omega) = \Phi(-\omega)$$

$$(b) \quad \int_0^{\infty} \Phi(\omega) d\omega = R(0)$$

$$(c) \quad \Phi(\omega) \geq 0$$

(28)

It may be shown (see Eqs. (1), (4), (26), and Ref. 26) that the Fourier series coefficients c_n and the power spectrum are related as follows.

$$\frac{1}{2} c_n^2 = \Phi(\omega_n) \Delta\omega, \quad \text{where } \Delta\omega = \frac{2\pi}{T}$$

which for $T \rightarrow \infty$,

$$\rightarrow \Phi(\omega) d\omega \quad (29)$$

Thus returning to the representation of Eq. (15) we can write

$$x(t) = \sum_{n=1}^{\infty} \cos(\omega_n t - \psi_n) \sqrt{2\Phi(\omega_n)\Delta\omega} \quad (30)$$

which in continuous form is sometimes written as

$$x(t) = \int_0^{\infty} \cos(\omega t - \psi) \sqrt{2\Phi(\omega) d\omega} \quad (31)$$

The continuous form of Eq. (30) looks somewhat odd but serves to indicate how a time history of $x(t)$ may be generated from a continuous spectrum by random choices

of phase angles. As the interval $\Delta\omega$ becomes small, the time functions approach those for the continuous spectrum.

(c) Relations Between Spectra and Time History Characteristics in the Gaussian Case

The foregoing has served to indicate how the power spectrum completely defines a stationary Gaussian random process. This would tend to imply that all statistical characteristics of the time history can be determined from the spectrum. While this is true, in principle, the expressions are exceedingly complicated, involving integrations in multi-dimensional spaces. As a result, simple and useful results have only been obtained for some of the more elementary questions of interest. Such problems are considered in some detail in Part III of Ref. 26. Many of the expressions that have been evaluated are, however, of practical use, and a few of these are noted here.

It will be recalled, Eq. (28), that the integral of the spectrum determines the mean square value. The root mean square value may thus be obtained from the spectrum and provides a useful linear measure of the disturbance intensity. If the disturbance $x(t)$ has a normal or Gaussian probability distribution with zero mean, the probability density $P(x)$ is given by

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/2\sigma^2} \quad (32)$$

The probability distribution defines the proportion of total time that the disturbance is at any value and is in this case completely specified by the root mean square value.

Rice (Ref. 26) has derived a number of relations which appear useful in aeronautical applications and are particularly significant for fatigue studies. Of these, the more

important appear to be expressions for the average number of times per second that a given value of x_1 is crossed with positive slope and the average number of peak values (maxima) per second that are above a given value x_1 of x . These quantities are indicated in Fig. IVC:2 below

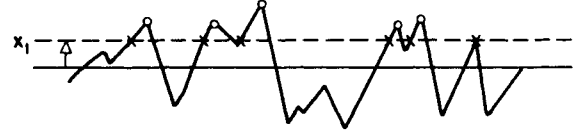


Fig. IVC:2

The average number of times per second that a given positive value x_1 of x is crossed with positive slope (or a given negative value x_1 of x is crossed with negative slope) is given by

$$N_c(x_1) = \frac{1}{2\pi} \frac{\sigma_1}{\sigma} e^{-x_1^2/2\sigma^2} \quad (33)$$

where σ_1 is the root mean square value of x' (the time derivative of $x(t)$) and is related to the spectrum of $x(t)$ by the relation

$$\sigma_1 = \sqrt{[x'(t)]^2} = \left[\int_0^\infty \omega^2 \Phi(\omega) d\omega \right]^{1/2} \quad (34)$$

The expression for the number of peaks (maxima) per second, $N_p(x_1)$, that are above a given value x_1 (or the average number of minima or "valleys" per second that are below $-x_1$) is complicated but for large values of y_1

$$N_p(x_1) \approx N_c(x_1). \quad (35)$$

For narrow spectra, this approximation

is good over the whole range of values of x_1 . For most other practical cases, it appears that this approximation is quite good for $x_1 > 2\sigma$. The average total number of peak (maxima) values of x_1 (or the number of minima) which includes the maxima for positive and for negative values of x_1 , is given simply by

$$N_p = \frac{1}{2\pi} \frac{\sigma_2}{\sigma_1} \quad (36)$$

where

$$\sqrt{\left[x''(t)\right]^2} = \left[\int_0^\infty \omega^4 \Phi(\omega) d\omega \right]^{\frac{1}{2}}$$

The foregoing relations between the spectrum and the peaks, Eq. (33) to (36), which apply to completely Gaussian disturbances, have been tested empirically in several aeronautical cases with success, as will be illustrated in the applications to be presented.

(d) Cross-Spectra

The generalization of power spectra which is already of substantial importance in aeronautics and whose importance can be expected to increase is the generalization to cross-spectra of two or more "noises." Here for continuous spectra we may write

$$\text{ave} \{x(t) y(t + \tau)\} = \frac{1}{2} \int_{-\infty}^{\infty} e^{i\omega\tau} \Phi_{xy}(\omega) d\omega \quad (37)$$

where $\Phi_{xy}(\omega) d\omega$ is the (complex) cross-spectrum, and its real and imaginary parts, defined by

$$\Phi_{xy}(\omega) = C_{xy}(\omega) + i q_{xy}(\omega) \quad (38)$$

where $c_{xy}(\omega)$ and $q_{xy}(\omega)$ are the cospectrum and the quadrature spectrum, respectively.

The basic symmetry conditions and inequalities satisfied by the cross-spectra are

$$(1) \quad c_{xy}(\omega) = c_{yx}(\omega) = c_{xy}(-\omega)$$

$$(2) \quad q_{xy}(\omega) = -q_{yx}(\omega) = -q_{xy}(-\omega)$$

$$(3) \quad \frac{[c_{xy}(\omega)]^2 + [q_{xy}(\omega)]^2}{\Phi_x(\omega) \Phi_y(\omega)} \leq 1$$

$$(4) \quad \text{ave} \{x(t) y(t)\} = \frac{1}{2} \int_{-\infty}^{\infty} \Phi_{xy}(\omega) d\omega \quad (39)$$

We may consider that the cospectrum $c_{xy}(\omega)$ and the quadrature spectrum $q_{xy}(\omega)$ represent the power of the in-phase and 90 degree out-of-phase components respectively of the two disturbances.

A number of engineering applications of cross-spectra will be indicated subsequently.

(e) Non-Gaussian Processes

The preceding discussion has been confined to the simplest case of Gaussian processes. For these processes, we noted that the process was essentially defined by the autocovariance functions and power spectrum. Although the autocovariance function and power spectrum apply in general, they no longer completely define the process in the non-Gaussian case. Third and higher order moment functions and higher order spectra appear required in these cases. Investigations of these higher order moments are only beginning, and some initial results have been given, for example, in Ref. 13.

(f) Input-Output Relations

Single Disturbance. A particularly useful and simple relation exists for linear systems between the spectrum of a disturbance and the spectrum of the system response to the disturbance.

From Eqs. (13) and (14) we may write

$$Y(\omega) = X(\omega) T(\omega)$$

which leads to

$$\Phi_o(\omega) = \Phi_i(\omega) |T(\omega)|^2 \quad (40)$$

where

$\Phi_o(\omega)$ is the output spectrum

$\Phi_i(\omega)$ is the input power spectrum

$T(\omega)$ is the frequency response function

and describes the system response to sinusoidal disturbances of various frequencies.

Eq. (40) indicates that the response at a given frequency depends only on the input and the system admittance characteristics at that frequency. This is readily plausible for linear systems. In the interest of concreteness, the application of this relation to the gust loads case is illustrated in Fig. IVC:3.

The top sketch on this figure is the input spectrum and in this case represents the spectrum of atmospheric vertical velocity. The frequency argument Ω , which is 2π divided by the gust wavelength, is introduced in place of ω because gust disturbances experienced by an airplane are essentially space disturbances rather than disturbances in time. The several measurements that

have been made to establish this spectrum will be reviewed later.

The second sketch $|T(\omega)|$ represents the squared amplitude of a specified airplane response, such as the airplane normal acceleration, to sinusoidal gusts of unit amplitude and of frequency Ω . The sketch shown is a typical normal acceleration response for a moderately flexible airplane. The peaks of the curve represent airplane modes of response such as the free body and fundamental wing bending modes illustrated. The output spectrum $\Phi_o(\Omega)$ is obtained in accordance with Eq. (40) as the product of the first two curves $\Phi_i(\Omega) \times |T(\Omega)|^2$ and gives the power spectrum of either normal acceleration or a relevant stress.

The lack of specific accounting for the phase in Eq. (40) for either the disturbance or the system response is an important simplification introduced by the spectral technique, and is frequently misunderstood. The reason for this condition may be indicated simply for the Gaussian case by returning to the Fourier series representation of Eq. (15).

$$x(t) = \sum_{n=0}^{\infty} c_n \cos(\omega_n t - \psi_n).$$

The response to this disturbance for a linear system can in turn be written as in the Fourier Series case Eq. (6)

$$y(t) = \sum_{n=0}^{\infty} c_n |T(\omega_n)| \cos(\omega_n t - \psi_n + \theta_n). \quad (41)$$

Inasmuch as the initial phase angles ψ_n are random and independent with a prescribed uniform probability distribution, the adjusted phase angle $(\theta_n - \psi_n)$ will likewise be random with the same probability distribution. Thus the output $y(t)$ is likewise a Gaussian process

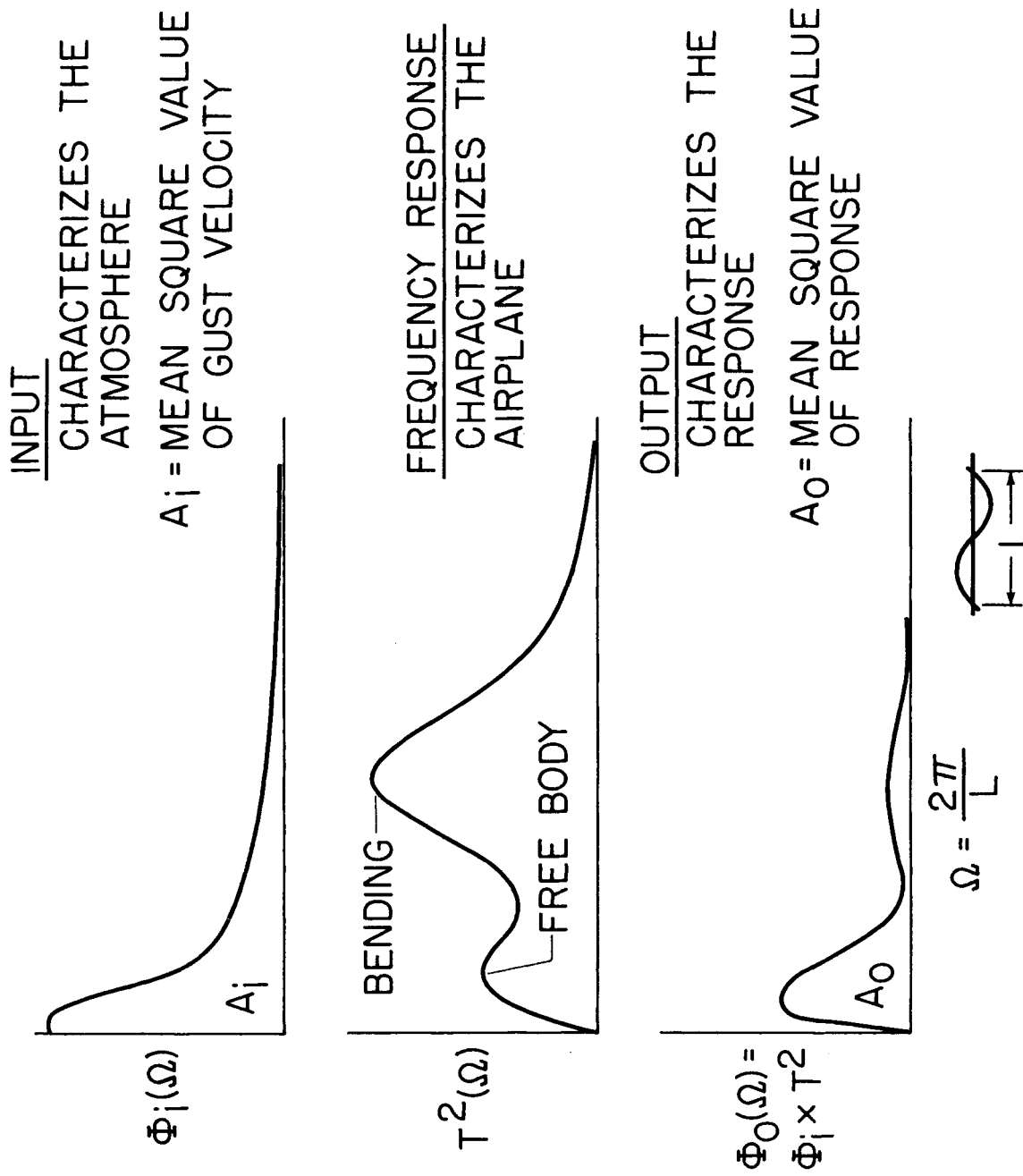


Figure IVC:3 Input-Output Relation for Power Spectra

and is also completely specified by the coefficients $[C_n |T(\omega)|]$.

By virtue of Eq. (29), we may write for $T \rightarrow \infty$ that

$$\frac{1}{2} (C_n |T(\omega)|)^2 \rightarrow \Phi_y(\omega) d\omega. \quad (42)$$

Thus, the output is likewise completely specified by the power spectrum.

Another useful relation for linear systems exists between the cross-spectrum of the input and output and the power spectrum of input. We note from Eqs. (13) and (14) that in the frequency plane we may write that

$$Y(\omega) = X(\omega) T(\omega). \quad (43)$$

This leads to the following relation between the spectrum of a disturbance and the cross-spectrum between the input and output:

$$\Phi_{xy}(\omega) = \Phi_x(\omega) T(\omega). \quad (44)$$

An important use of this equation is its application in the determination of both the amplitude and phase of the frequency response function from simultaneous time-history measurements of the disturbance and the system response.

Multiple Disturbances. It is worth noting that the response of a system to several simultaneous random disturbances such as vertical and longitudinal turbulence may also be handled in an analogous manner to that for the single disturbance.

For the case of two disturbances, the

response $y(t)$ is given by

$$y(t) = \int_0^\infty x_1(t_1) A_1(t-t_1) dt_1 + \int_0^\infty x_2(t_1) A_2(t-t_1) dt_1 \quad (45)$$

where $x_1(t)$ and $x_2(t)$ are the two disturbances and $A_1(t)$ and $A_2(t)$ are the unit impulse responses of the system to the respective disturbances.

In terms of the power spectra, the output is then given by

$$\begin{aligned} \Phi_y(\omega) &= \Phi_{x_1}(\omega) |T_1(\omega)|^2 + \Phi_{x_2}(\omega) |T_2(\omega)|^2 + \\ &\quad \Phi_{x_1}(\omega)_{x_2} T_1^*(\omega) T_2(\omega) + \Phi_{x_2}(\omega)_{x_1} T_1(\omega) T_2^*(\omega) \end{aligned} \quad (46)$$

where $\Phi_{x_1}(\omega)$ and $\Phi_{x_2}(\omega)$ are the power spectra of x_1 and x_2 .

$T_1(\omega)$ and $T_2(\omega)$ are the system frequency response functions for the respective disturbances (the asterisk designates the complex conjugate), and $\Phi_{x_1}(\omega)_{x_2}$ and $\Phi_{x_2}(\omega)_{x_1}$ are the cross-spectra which for stationary processes are defined by the following relations

$$\Phi_{x_1}(\omega)_{x_2} = \frac{1}{\pi} \int_{-\infty}^{\infty} R_{x_1 x_2}(\tau) e^{-i\omega\tau} d\tau \quad (47)$$

where

$$R_{x_1 x_2}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x_1(t) x_2(t+\tau) dt. \quad (48)$$

The cross-spectrum in contrast to the power spectrum, it will be recalled, is complex with the real part providing a measure of the in-phase power called the

co-power and the imaginary part providing a measure of the out-of-phase power, called the quadrature power.

Eq. (44) may be used directly to determine the spectrum of the output to two random disturbances. For purposes of determining the spectrum of the input disturbances from measurements of the output response, measurements are required in this case of an additional output response,

$$z(t) = \int_0^{\infty} x_1(t_1) A_3(t-t_1) dt_1 + \int_0^{\infty} x_2(t_1) A_4(t_1-t) dt_1$$

where A_3 and A_4 are the unit impulse responses for the output quantity Z for the respective disturbances.

Measurements of the spectra of y and z and their cross-spectra may be used to obtain the input spectra and cross-spectra. Such applications have, for example, been made in aeronautics and in the determination of two (or more) components of turbulence affecting an airplane in flight (see, for example, Ref. 9). A further generalization to multidimensional disturbances is relatively straightforward and is already important in turbulence, ocean wave, and buffeting studies (see, for example, Refs. 14 and 23).

For convenience, the principal aspects of the various techniques discussed for response determination for both arbitrary and random disturbances are summarized in the table at the bottom of the page.

COMPARISON OF VARIOUS TECHNIQUES FOR RESPONSE DETERMINATION

Type Function	Technique	Representation of Input	Transfer Function	Input-Output Relation	Nature of Output Information
Recurring	Fourier Series	$\sum_1^{\infty} c_n \cos(\omega_n t - \psi_n)$	$T(\omega_n)$	$\sum_1^{\infty} c_n T(\omega_n) \cos(\omega_n t - \psi_n + \theta_n)$	Time history
Dissipating	Fourier Integral	$\frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega$	$T(\omega)$	$\frac{1}{2\pi} \int_{-\infty}^{\infty} T(\omega) X(\omega) e^{i\omega t} d\omega$	Time history
Arbitrary	Impulse Response	$x(t)$	$A'(t)$, unit impulse response	$\int_0^{\infty} x(\tau) A'(t-\tau) d\tau$	Time history
Random	Generalized Harmonic Analysis	Power spectrum, $\Phi(\omega)$ Limit $\frac{1}{T} \int_0^T X_T(\omega) ^2 d\omega$	$ T(\omega) ^2$	$\Phi_y(\omega) = T(\omega) ^2 \Phi_x(\omega)$ $\Phi_{xy}(\omega) = T(\omega) \Phi_x(\omega)$	Power spectrum & related statistical characteristics

(g) Significance of Power Spectral Techniques

The foregoing discussion has served to indicate the basic concepts and relations of the power spectral and random process techniques. These results have some far-reaching implications in regard to aeronautical problems involving random disturbances. They provide a systematic mathematical framework and technique for the analysis of such problems which formerly were treated by empirical and intuitive methods which were frequently arbitrary.

Some of the more important implications that should be noted are the following:

(1) The power spectrum provides a compact mathematical representation of a random process.

(2) Relative to random disturbances, the significant characteristics of a linear dynamic system are contained in the frequency response function.

(3) The significant characteristics of the response of linear systems to random disturbances may be calculated simply (although the calculated information is only complete in the case of a Gaussian disturbance).

(4) A need exists for detailed information on the spectral characteristics of the random processes that occur in aeronautical applications.

(5) These analytical techniques provide a means of obtaining results that can be directly correlated with test data for random type disturbances.

(6) Response to multiple disturbances, taking into account interrelations between the disturbances, may be obtained in a systematic fashion.

(7) Perhaps the most important result offered by these techniques is the

new insight into the dynamic interrelations between disturbances and system responses. This new insight provides a valuable guide in planning new theoretical and experimental investigations.

(8) Information on the random type input may be obtained readily from a knowledge of the system frequency response characteristics and measurements of the system response.

(9) The significance of the effects of changes in the frequency response function on the response to a random disturbance can be evaluated in a systematic fashion.

IVC:4 THE MEASUREMENT OF POWER SPECTRA

From the foregoing, it is clear that the power spectrum plays a central role in the analysis of problems involving random disturbances. Not only can the spectrum of a stationary random process $x(t)$ be defined in two equivalent ways, namely:

$$\phi(\omega) = \frac{2}{\pi} \int_0^{\infty} R(\tau) \cos \omega \tau d\tau \quad (49)$$

where

$$R(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) x(t + \tau) dt$$

and

$$\phi(\omega) = \lim_{T \rightarrow \infty} \frac{1}{\pi T} |X_T(\omega)|^2 \quad (50)$$

where

$$X_T(\omega) = \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) e^{i\omega t} dt. \quad (51)$$

but it can also be defined in terms of the mean-square fluctuation of the result of filtering the noise.

If the process is passed through a filter of frequency response function $T(\omega)$, the mean square fluctuation is

$$\int \phi_0(\omega) d\omega = \int |T(\omega)|^2 \phi(\omega) d\omega \quad (52)$$

and this implicitly defines $\phi(\omega)$ in terms of such mean square fluctuations.

The choice of the relation to be used in practice is a question of computational engineering. For digital calculations, Eq. (49) appears preferable while for analog techniques, Eq. (52) is preferable, for reasons to be indicated.

We shall here discuss some of the considerations involved in the measurement of power spectra. Based on these considerations, a recommended procedure for digital calculations is outlined in Appendix A. A few remarks on analog computers are also given in Appendix B. Crude, rapid, digital methods are discussed in Appendix C.

In both Eqs. (49) and (50), the power spectrum is defined in terms of infinite integrals. In practice, the information available consists of a piece of $x(t)$ and frequently a very short piece. Thus, the infinite integrals of Eqs. (49) to (50) (crude digital methods may be better based on other formulas) must be approximated. In addition, if the calculations are to be performed numerically, the evaluations must further be restricted to values at uniformly spaced time intervals.

These finite approximations have a number of serious consequences in regard to the accuracy and distortion of the true spectrum. We shall next briefly discuss some of these effects.

(a) Aliasing

In numerical calculations, we are frequently concerned with data consisting of values of $x(t)$ at equally spaced intervals of time. Such systematic sampling alters the spectrum in an important way which is often overlooked. The important observation is this: "A fast cosine wave systematically sampled at regular intervals appears the same as a low cosine wave." See Fig. IVC:4.

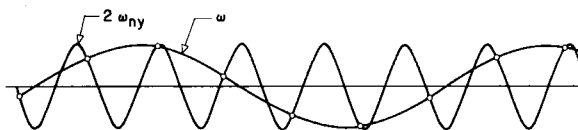


Fig. IVC:4

Thus, after sampling, the apparent content at any frequency is the result of superposing the content of many frequencies, which have now become aliases of one another. If we define the Nyquist, angular frequency ω_{Ny} by

$$\omega_{Ny} = \frac{\pi}{\text{interval between samples}} \quad (53)$$

then the following angular frequencies

$$\omega, 2\omega_{Ny} \pm \omega, 4\omega_{Ny} \pm \omega, \dots$$

are confounded and become aliases of one another.

This fact has substantial repercussions, both in the interpretation of the analysis and in the design of equipment. In order to avoid difficulties from this source, the interval selected for record reading must be small enough to insure that no appreciable power exists above the corresponding Nyquist frequency. In practice, this may not always be practicable, and consideration must instead be given to removal of the power at the higher frequencies either by rejection in the measuring instruments or by filtering the recorded measurements.

(b) Limited Resolution

The finite length of the record inevitably leads to finite resolving power in frequency, and this means that in practice we can only measure average power over frequency bands of finite widths. This averaging leads to a "smearing" or distortion which may be conveniently viewed as equivalent to convolving the true spectrum with a band-pass filter as given by the following relation:

$$\Phi_{\text{meas}}(\omega) = \int_{-\infty}^{\infty} \Phi_{\text{true}}(\omega_1) A(\omega - \omega_1) d\omega_1 \quad (54)$$

where $A(\omega)$ is the filter or "spectral window." Thus, measurements can be considered to provide measurements of average power over more or less wide frequency intervals.

Ideal "square" filters are impossible to achieve and difficult to approximate either electrically or in numerical calculations. Narrow filters, as we shall see below, have serious drawbacks. The simplest electrical circuits yield filters with a large and roughly triangular transmission region about the center frequency but which have finite (though small) transmission at all frequencies. The effects of these small skirts is frequently troublesome. (The shoulders may be "squared up" considerably without difficulty, but this is not too useful.)

Numerical calculations may also be tailored to yield a filter which is roughly a large triangle, but which also contains ripples or side lobes at both adjacent and distant frequencies. Obviously the choice of an appropriate filter or "window" plays an important part in spectral measurement and in large measure determines the distortion of the true spectrum due to "diffusion" or "leakage" of power from adjacent or distant frequencies. The choice of a spectral window will, of course, be particularly important for non-uniform spectra and critically important if large peaks are

present. This problem will be considered further at a later point.

(c) Sampling Errors

Sampling errors are the errors resulting from the necessary observation of only a small part of the ensemble of functions of interest. We shall only sketch the broad picture here. The details may be found in Ref. 25. If we consider non-overlapping bands of frequencies narrow enough for $\Phi(\omega)$ to be relatively constant, the input variability to be expected in the Gaussian case can be expressed conveniently in terms of the statistical coefficient of variation.

If we estimate

$$\Delta\Phi = \int_{\omega_1}^{\omega_2} \Phi(\omega) d\omega$$

then the coefficient of variation for $\Delta\Phi$ defined by

$$\frac{\text{root mean square deviation of } \Delta\Phi}{\text{average of } \Delta\Phi}$$

can be shown from the results of Ref. 25 to be given approximately by

$$\frac{1}{2\pi} [(\text{length of record}) (\omega_2 - \omega_1)]^{-\frac{1}{2}} \quad (55)$$

The root mean square deviation provides a useful measure of the statistical reliability of measurements and may be used to determine confidence bands (bands within which the average value may be expected to be with a given probability). If we substitute the measured value as a rough estimate of the average value, the interval defined by

$$\Delta\Phi \pm k \text{ RMS}_{\Delta\Phi}$$

which is equal to

$$\Delta\Phi \pm k \frac{1}{2\pi} [(\text{length of record}) (\omega_2 - \omega_1)]^{-\frac{1}{2}}$$

provides confidence bands for a given probability level depending upon the value of k .

If the coefficient of variation is small, say less than 0.2, the estimate of $\Delta\Phi$ is approximately normally distributed. For this case, the probability levels for several values of k are:

k	Probability Level
1.0	0.68
1.65	.90
1.96	.95
2.58	.99

If the coefficient of variation is not small, this normal distribution approximation is not very good and a more exact procedure, described in Ref. 25, has to be used.

In the foregoing, the true spectrum has been assumed to be essentially constant over the interval of estimation. If the spectrum varies appreciably over the filter band width, the statistical reliability of the estimates is reduced as indicated in Ref. 25. This loss of reliability is one of the many difficulties involved in the measurement of spectra having large variations in density.

In practice, estimates of average spectral densities do not refer to averages of the simple sort discussed above. Filters, either analog or digital, in practice as indicated earlier are of the form

$$\int A(\omega)\Phi(\omega) d\omega$$

where the window $A(\omega)$ generally consists mainly of a large smooth and roughly triangular central bulge surrounded by other minor lobes or ripples in the numerical case or continuous side bands in the analog case.

For the accuracy required by our present purposes, we can replace such a window

by a rectangular window approximately equal in width to the separation $\Delta\omega = \omega_2 - \omega_1$, where ω_2 and ω_1 are the frequencies at which the transmission is one-half the peak value (roughly one-half the base of the triangular central bulge of the filter).

It is clear from the foregoing that the sampling variability as measured by the coefficient of variation is approximately inversely proportional to the square root of the record length and the square root of effective width of the spectral window. Thus, increasing the width of the spectral window increases the statistical reliability of the estimates but at the expense of greater distortion. Alternatively, narrowing the spectral window decreases the statistical reliability. In practice, a wise compromise is required between these two conflicting considerations.

Little is known about the sampling variability of cross-spectra estimates and this problem is currently under study. Cross-spectra are frequently used, as suggested by Eq. (44) in the determination of the frequency response function, viz.

$$T(\omega) = \frac{\Phi_{xy}(\omega)}{\Phi_x(\omega)} \quad (56)$$

If both the spectrum and the cross-spectrum are calculated from the same record samples, the system frequency response function will frequently be well-determined even though the sampling variability of the separate spectra is large. This can occur because the statistical fluctuations in the spectrum and cross-spectrum are not independent.

While it would be possible, if the memory time of the system were short compared to the lengths of the available records, to approximate the transfer function using the general expression for the frequency response function for an arbitrary disturbance

starting at time zero

$$T(\omega) = \frac{\int_0^{\infty} y(t) e^{-i\omega t} dt}{\int_0^{\infty} x(t) e^{-i\omega t} dt} \quad (57)$$

the computational labor involved would be substantially greater, and the accuracy of the result would not be appreciably improved (and might be worsened).

(d) Other Errors

Mention should also be made of the effects of instrument, reading, and computational errors. The effects of such error propagation are already critical in many applications and form the motivation for the wide attention being given to filtering and data smoothing techniques. The character of these errors will differ widely between cases and may in some cases be critical for spectra determinations.

These errors may frequently be kept reasonably small, of the order of one percent, and tend to be more or less constant with frequency. If the spectrum being measured is reasonably flat, no serious difficulties may be encountered from these sources. However, if the spectrum has a wide range of intensity, the effect of these errors will be small in regions of peak power but may provide a sizeable, if not the major, portion of the power in the lower power regions of the spectrum, thus almost completely hiding the true spectrum from our measurements.

Possible ways of avoiding such difficulties are mentioned below at the close of the discussion of prewhitening.

(e) Spectral Windows

We have seen that the choice of the spectral window has important effects on the distortion and reliability of the spectrum estimates.

In the numerical calculation, the approximation for the autocovariance function used has important effects on the shape of the spectral window.

We can see this clearly if we take

$$R_{\text{meas}}(\tau) = R_{\text{true}}(\tau) \alpha(\tau)$$

where $\alpha(\tau)$ is a weighting function, we obtain

$$\Phi_{\text{meas}}(\omega) = \int_{-\infty}^{\infty} \Phi_{\text{true}}(\omega_1) A(\omega - \omega_1) d\omega_1$$

where the spectral window is given by

$$A(\omega) = \int_{-\infty}^{\infty} \alpha(\tau) e^{-i\omega\tau} d\tau$$

If we simply truncate the autocovariance where

$$\alpha(\tau) = \begin{cases} 1 & -\tau_0 \leq \tau \leq \tau_0 \\ 0 & \text{elsewhere} \end{cases}$$

the following is obtained

$$\Phi_{\text{meas}}(\omega) = \int_{-\infty}^{\infty} \Phi_{\text{true}}(\omega_1) \tau_0 \frac{\sin \tau_0(\omega - \omega_1)}{\tau_0(\omega - \omega_1)} d\omega_1$$

where our spectral window is of the $(\sin x)/x$ type as shown in the sketch with a sizeable peak at $\omega = 0$ and smaller, although appreciable side lobes at $\omega = \pm (3/4)(\pi/\tau_0)$, $(5/4)(\pi/\tau_0)$, etc.

Clearly, this is not a very satisfactory spectral window. We note that the smaller the value chosen for τ_0 , the wider the base of the main peak, a condition favoring statistical reliability, but increasing the amount of distortion

A significant improvement in the spectral window may be achieved quite simply by an appropriate choice for the weighting function $\alpha(\tau)$. (For numerical calculations, the effects of the discrete data must also be taken into account. Blackman in Ref. 30

has shown how the analysis for this case can conveniently be simplified by the use of a "Dirac comb.")

We shall pursue this matter no further here, but merely note that the calculation procedure for digital computations outlined in Appendix A incorporates a simple but quite effective modification of this function of the form indicated in Fig. IVC:5. As indicated in the appendix, the spectral window is roughly triangular with a base width that may be varied in accordance with the dictates of diffusion and statistical reliability consideration. The side lobes (or ripples) are small, ± 2 percent of the center peak value, but sometimes troublesome if the spectrum contains sharp peaks. The direct improvement of this side lobe condition appears extremely difficult and troublesome. An alternative solution is proposed in the next section.

It should also be noted that the use of the truncated autocovariance function in accordance with the above discussion greatly reduces the amount of calculation without a loss of significant information. This reduction in the amount of calculation provides good reason for the choice of Eq. (49) over Eq. (50) in digital computations. (If

only a few estimates are to be made, Eq. (52) could be used in digital computation, but when the usual number of estimates are required Eq. (49) will be preferred.)

(f) Prewhitening

Because the errors and distortion introduced in the computation are often rather constant with frequency, the least pressing demands on accuracy will be made by moderately white (that is, pastel) noise spectra. Such noises will have statistical uncertainties in estimated average spectra densities of about the same size at all frequencies.

Demands on accuracy and, conversely, degradation of results from computational noise and distortion may often be reduced by analyzing the spectrum of a transformed noise, whose spectrum has been altered to a known extent and in such a direction as to make its spectrum more nearly white. Such a transformation will be called prewhitening. (It is, of course, understood that when prewhitening is used, the resulting estimates of averaged values of $R(\tau)$ or $\Phi(\omega)$ are to be adjusted in the course of the computation in a way compensating for this transformation.)

Another important consideration also points toward prewhitening. Minor lobes in digital filters and gradual cutoffs in analog filters always exist, and are to be reduced only with difficulty. In the computational filters involved in the computation of estimated average values of $\Phi(\omega)$, this fact is somewhat uncomfortable, since we would prefer averages over short, clearly restricted frequency bands. Such difficulties with "leakage" or "diffusion" of power are least for a flat spectrum and increase as the possibility increases of a high spectral peak leaking through a minor lobe and contributing an amount comparable with that coming from a low portion of the spectrum under the major lobe.

Two methods of prewhitening are available once the input is at hand, the two familiar

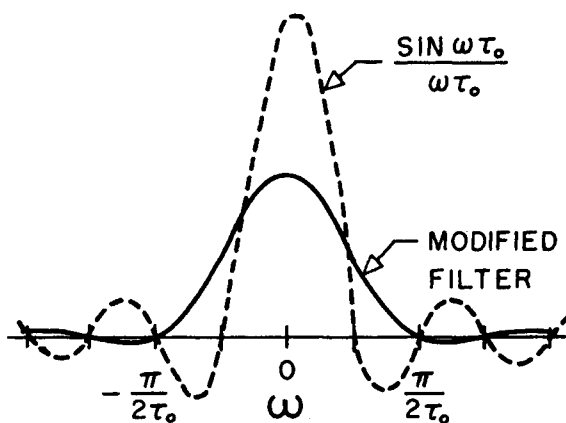


Fig. IVC:5

methods of linear filtering. If we are dealing with an analog signal, we may use an ordinary analog filter. If we are dealing with equispaced digital data, we use a corresponding digital filter.

If $L(\omega)$ is the transfer function of the filter, expressing amplitude and phase change as a function of frequency, formally the filter action being

$$e^{i\omega t} \rightarrow L(\omega) e^{i\omega t}$$

then the effect on the spectrum is

$$\Phi(\omega) \rightarrow |L(\omega)|^2 \Phi(\omega) = L(-\omega) L(\omega) \Phi(\omega)$$

and we must compensate our estimate of $\Phi(\omega)$ by division by $L(-\omega) L(\omega)$.

The determination of $L(\omega)$ for particular analog filters and the synthesis of analog filters with prescribed transfer functions are familiar operations. The corresponding matters for the digital equispaced case are less well-known and deserve a brief mention.

The simplest nontrivial such filter is the transformation

$$y(t) = x(t) - \theta x(t-1)$$

If we place $x(t) = e^{i\omega t}$ we find immediately that

$$L(\omega) = 1 - \theta e^{i\omega}$$

$$L(-\omega) L(\omega) = 1 + \theta^2 - 2\theta \cos \omega.$$

It is easy to see that transformations of the form

$$y(t) = a_0 x(t) + a_1 x(t-1) + \dots + a_n x(t-n)$$

will produce a polynomial of degree n in $\cos \omega$ for $L(-\omega) L(\omega)$ and that any such real and non-negative polynomial can be obtained by a proper choice of the coefficients $a_0, a_1, \dots, a_n$.

The inverse transformations to these "moving average schemes" are often called autoregressive schemes, and are occasionally quite useful, either alone or in combination with moving average schemes; make $L(-\omega) L(\omega)$ the reciprocal a polynomial in $\cos \omega$. In combination, moving average and autoregressive schemes make possible the attainment of any rational function of $\cos \omega$ as an $L(-\omega) L(\omega)$.

The simplest autoregressive scheme is the inverse of $x(t) = y(t) - \theta y(t-1)$, namely

$$y(t) = \theta y(t-1) + x(t)$$

where, as for a general autoregressive scheme, successive values of $y(t)$ can be calculated from the values of $x(t)$ and the previous calculated values of $y(t)$.

When dealing with two or more series simultaneously, it appears, as a general rule, best to prewhiten each series separately and then use these prewhitened series for all spectra and cross-spectra.

Finally, prewhitening by other means sometimes offers a way to avoid the difficulty with recording and reading errors mentioned above. If the prewhitening can be done before the record is made, for instance by appropriately filtering or accelerometer output before recording it, then the recorded (and read) spectrum will be nearly flat, and difficulty with errors (which are likely to have an approximately white spectrum) will be minimized. This is often most desirable.

IVC:5 APPLICATIONS

In this section, some of the recent results obtained in the applications of power spectral techniques will be indicated. The results to be presented are not new, but are included in order to provide some concrete examples of the types of applications so far made and the nature of the results.

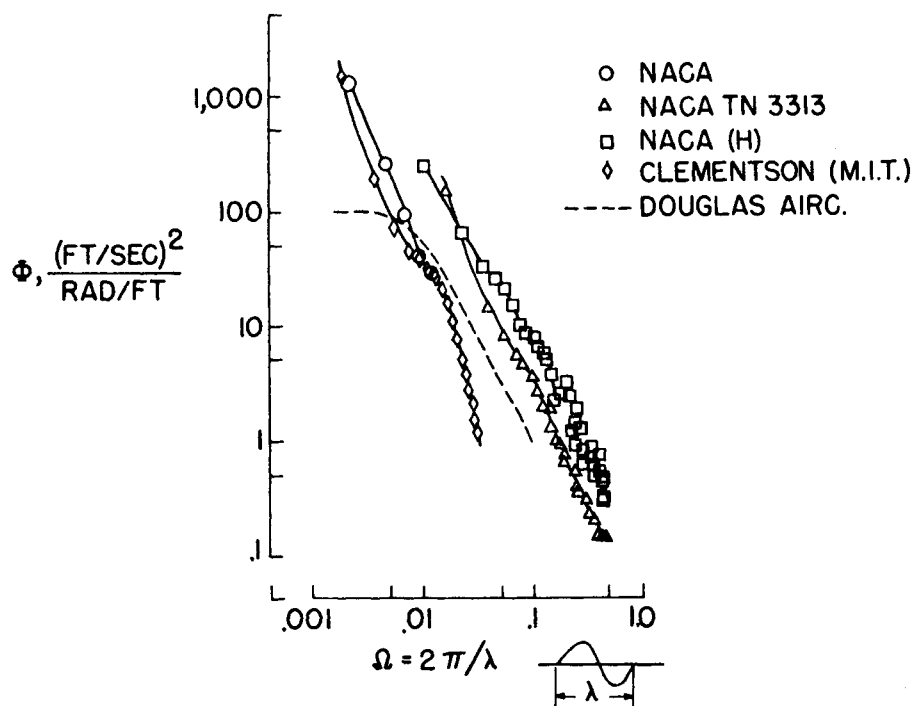


Fig. IVC:6 Measured Power Spectra of Atmospheric Turbulence

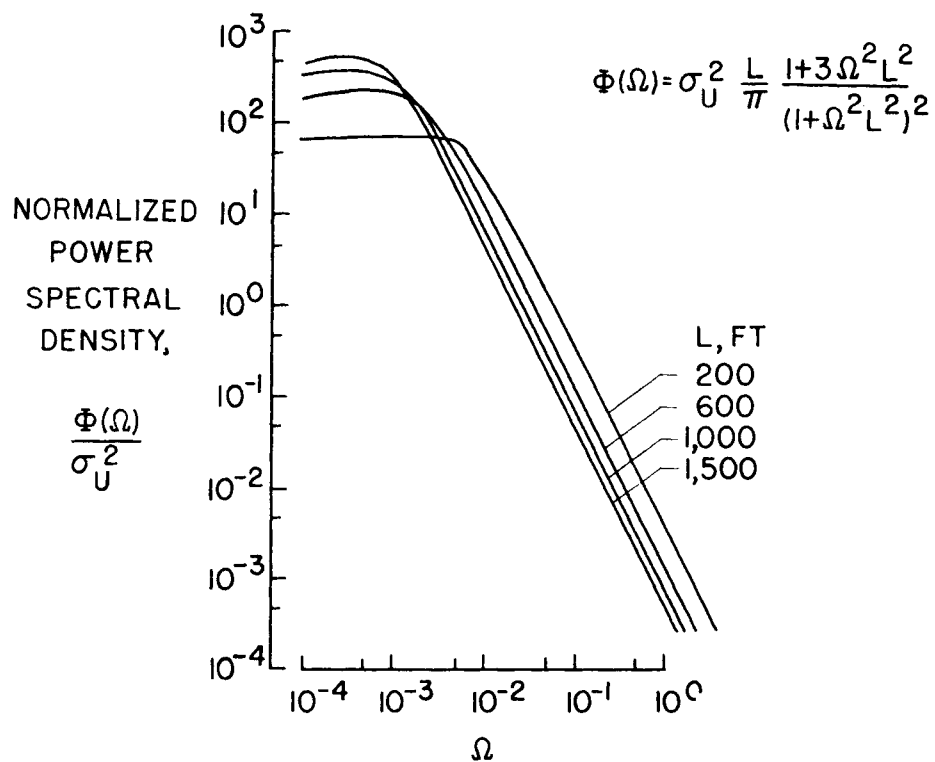


Fig. IVC:7 Analytic Representation of Atmospheric Turbulence Spectra

Many of the early applications have been concerned with the measurement of the characteristics and particularly the power spectra of various random processes of concern and some of these spectra measurements will be presented. Also, some results will be presented on the effects of short period damping on gust loads, the effects of wing bending flexibility on airplane bending moments and structural stress in rough air, and on buffeting loads.

(a) Disturbance Characteristics

The relatively complete theory existing for the case of a Gaussian disturbance, as has been indicated, has made this type of random process of particular interest. Thus, in many studies, efforts have been made to determine the degree to which the actual random processes encountered in aeronautical problems can be represented by a Gaussian process. In almost all cases, testing of observed measurements has been restricted to the consideration of the overall probability distribution.

It will be recalled that a Gaussian process is characterized by much more stringent conditions; namely, that the multivariate distribution be Gaussian. However, the indications of the overall probability distribution, while inconclusive, are a very helpful guide in most cases. Fortunately, in most of the applications so far made, the processes observed appear to be Gaussian to a good degree of approximation. Some of the quantities which appear to be so characterized at least for some conditions are atmospheric turbulence, airplane loads in rough air, buffeting loads, ocean wave heights, and pressure fluctuations in acoustic noise. Some experimental data in this regard are given in Refs. 5, 8, and 9.

Spectra of Atmospheric Turbulence. As an example of the results obtained in experimental measurements of power spectra,

some measurements of the spectrum of atmospheric turbulence will be described. A number of flight measurements of turbulence spectra have recently been made using airplanes, and some of these results obtained from various sources are summarized in Fig. IVC:6.

The ordinate of Fig. IVC:6 is the power spectral density, the abscissa is frequency in radians per foot. The frequency argument $\Omega = 2\pi/\lambda$ where λ is the gust wavelength, is useful for airplane studies since insofar as the airplane is concerned, turbulence is essentially a spatial phenomenon since the movement of "eddies" is slow compared to the motion of the airplane.

The data shown, in most cases, are for the vertical component of the turbulence. In one case, marked by the squares, the data are for the horizontal component or longitudinal turbulence. The data cover a range of gust wavelengths of from 10 to 3000 feet, which is the region that is of most concern for many airplane problems.

The spectra differ in intensity as indicated by the height of the spectra reflecting the variation of turbulence intensity with the weather conditions. The spectra, in each case, show rapid decreases in power with increasing frequency. In fact, the spectra vary roughly as $1/\Omega^2$, a result in general agreement with theoretical results obtained in the theory of isotropic turbulence. The general shape of these spectra, with rapid decreases in power at the higher frequencies, leads to some important implications in regard to the design of airplanes for smoother flight.

Because of the general characteristics of these spectra, it has been convenient in theoretical studies to use the analytical expression for the turbulence spectrum shown on Fig. IVC:7. This expression has been useful in wind tunnel studies of turbulence and can be seen to have the general characteristics of the measured spectra of atmospheric turbulence.

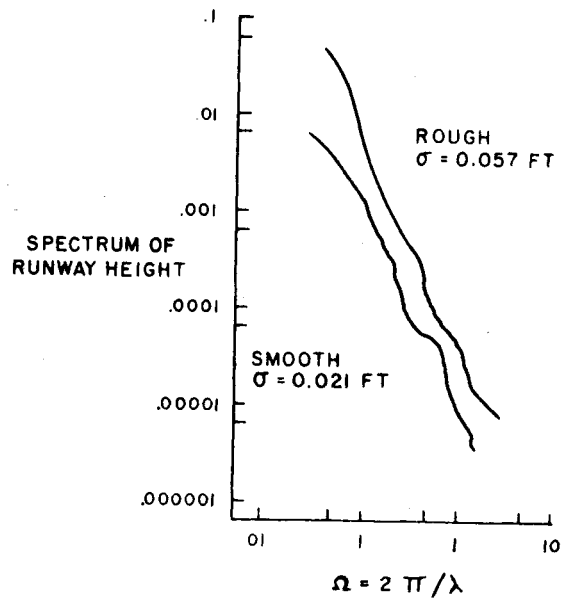


Fig. IVC:8 Measured Power Spectra of Runway Roughness

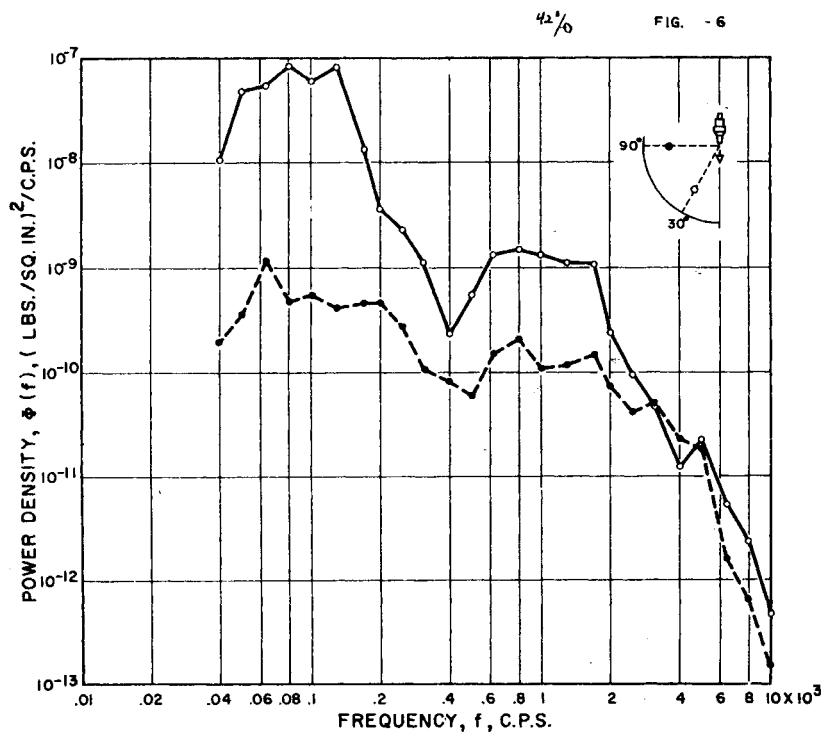


Fig. IVC:9 Measured Power Spectra of Noise from Jet Engine at a Distance of 200 Ft.

The equation has two parameters, the mean square gust velocity $\overline{U}^2$, which describes the turbulence intensity, and the so-called scale of turbulence L . The scale of turbulence L can, in a rough sense, be considered to be proportional to the average eddy size. The curves in Fig. IVC:7 are for a mean square gust velocity equal to 1 and for values of $L = 200, 600, 1000$, and 1500 feet. The curves at higher frequencies all approach a shape of $1/\Omega^2$, but differ in the frequency at which the flattening out occurs. For large values of L , most of the power is at the very low frequencies.

Comparison of these curves with the measurements of Fig. IVC:6 and other measurements of the spectrum of atmospheric turbulence have suggested that a representative value of L for atmospheric turbulence is of the order of a thousand feet. For studies of wind tunnel turbulence, L may be expected to be much smaller, perhaps on the order of an inch.

For design purposes, the overall gust experience in operations is of concern. Presumably, the overall experience consists of various exposure times to each of the spectra of Fig. IVC:6 as well as for other spectra of turbulence associated with different weather conditions. The available spectra measurements have suggested that, as a first approximation, the spectral shape might be considered fixed and only the intensity or root-mean-square gust velocity considered to vary. Thus, the overall gust experience may be considered to consist of exposure to turbulence of variable root-mean-square gust velocity, but invariant spectral shape.

On this basis, operational data on gust loads have been used in Refs. 18 and 19 to estimate the probability distribution of the root-mean-square gust velocity for various types of airplane operations and for various weather conditions. These results provide a basis for the calculation of the airplane gust response experience in operations.

Spectra measurements have also been made in regard to other aeronautical problems. Fig. IVC:8 taken from Ref. 16 shows some results that have been obtained in regard to runway roughness. The figure shows the spectra that have been obtained for sections of two runways at Langley Air Force Base, Virginia. One of these runways was considered, as designated on the figure, a relatively smooth runway while the other runway was considered a relatively rough runway and was, in fact, no longer in use. Further measurements of the runway spectra are currently underway.

Fig. IVC:9 shows the results of some measurements obtained on the noise from a jet engine (reported in Ref. 21). The figure shows the power spectra of pressure fluctuation as measured at two positions in the sound field, as indicated by the sketch on the upper right hand side of the figure. A large number of such measurements have been made in order to define the noise field about jet engines and about propeller-driven aircraft.

These measurements of disturbance spectra have provided at least an initial basis for the calculation of various airplane responses; for example, the turbulence spectra have been useful in studies of airplane gust loadings, runway spectra have been useful for studies of airplane taxiing loads and the noise spectra have been useful in the study of the noise nuisance and in the study of fatigue failures of panels due to noise.

(b) The Effect of Low Damping in Short Period on Loads in Rough Air

One of the first analytical results obtained in the application of power spectral techniques to the gust load problem was the indication of the importance of the short period damping on the loads in rough air, Ref. 5. Experimental verification of this result was obtained in a test in which a rocket-propelled model was flown through

rough air at low altitude at high speeds (Ref. 8). The model covered a range of Mach numbers from about 0.60 to 1.00 which also included a fairly large range of longitudinal stability conditions. The period and cycles to damp to one-tenth amplitude of the short period oscillation for the test model is shown as a function of Mach number in Fig. IVC:10 and indicates a fairly rapid deterioration of the short period damping above a Mach number of about 0.9.

The test model was a tailless missile and was fired through rough air at the NACA test center at Wallops Island, Virginia. The total missile flight lasted only about 20 seconds and yielded 2 seconds of usable record at each of eight test speeds. Inasmuch as the samples that could be obtained were necessarily small, there was some doubt that statistically significant test results could be obtained with this technique. Fig. IVC:11 shows sample time histories of the missile normal (vertical) acceleration at two of the test Mach numbers which correspond to airspeeds of roughly 750 miles per hour and 600 miles per hour.

Theory had predicted pronounced differences in loads for these two test conditions and this can be seen qualitatively to be the case from the relative magnitude of the fluctuations. The missile vertical acceleration in both cases is seen to be somewhat

irregular although the predominating motion is seen to occur at a frequency of roughly 10 cycles per second. The frequency characteristics of the records will be considered in more detail in connection with the power spectra.

The frequency distributions of the acceleration amplitude were determined from the records (readings at 0.01 second) and are shown in Fig. IVC:12 for several test speeds or Mach numbers. The dashed lines are fitted normal distributions. The reasonable fit of the normal distribution curves suggests that the process may be considered to have a Gaussian amplitude distribution.

In Fig. IVC:13, some of the expected and measured power spectra are compared. The calculated spectra for three of the test Mach numbers are shown on the upper left and are based on airplane measurements of the turbulence over the test course and missile response theory. These results, as can be seen from the relative areas or the root-mean-square values shown, suggest rather rapid increases in loads for the speed range covered. Also, the spectra move progressively to higher frequencies with increasing speed and become more peaked thus closely reflecting the short period characteristics.

The measured spectra are shown on the lower right and were obtained by the digital

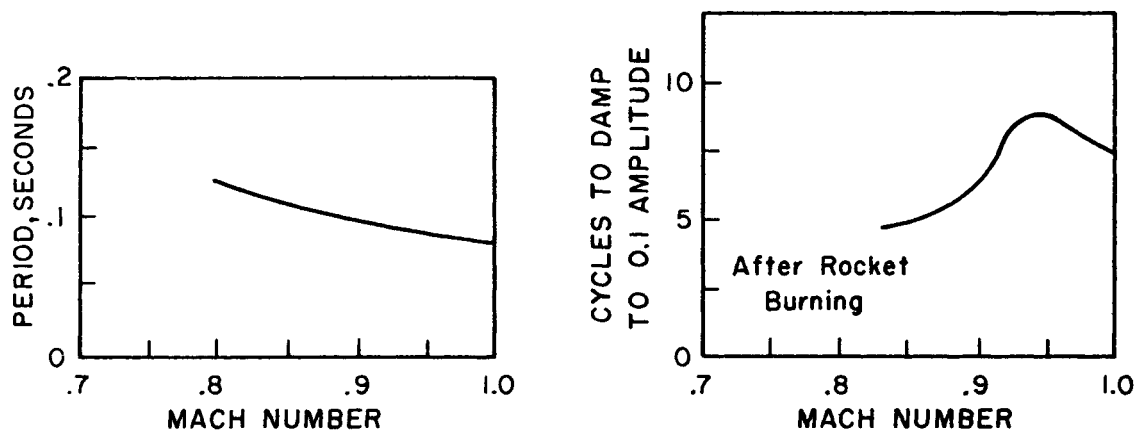


Fig. IVC:10 Missile Period and Damping

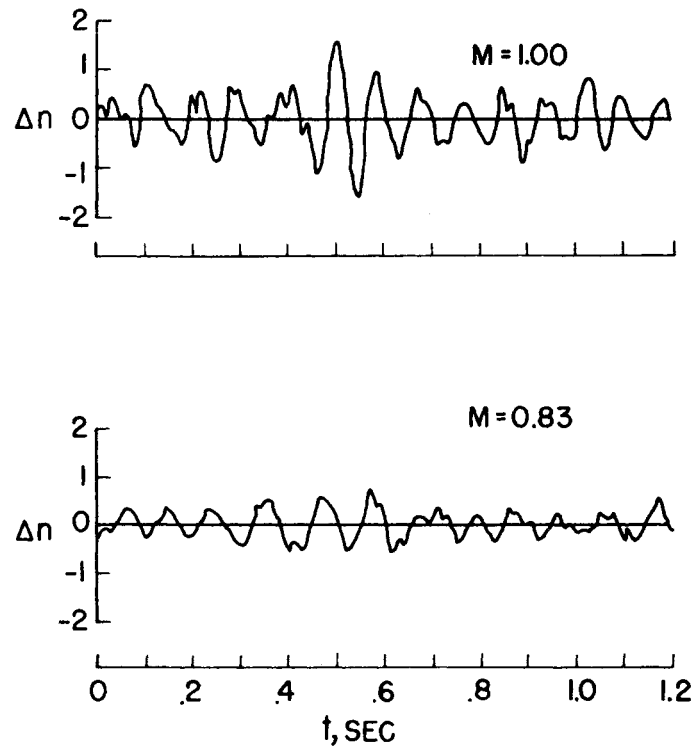


Fig. IVC:11 Time Histories of Missile Acceleration Response in Rough Air at Two Mach Numbers

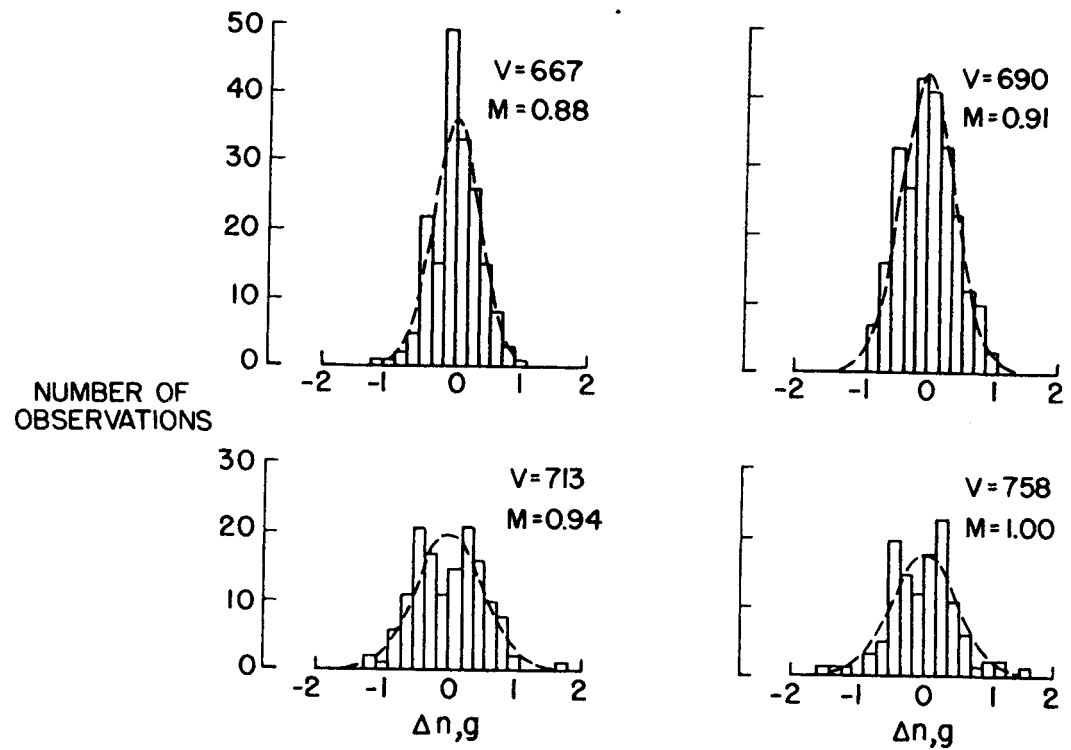


Fig. IVC:12 Measured Frequency Distribution of Missile Acceleration Response in Rough Air

SPECTRA OF MISSILE LOADS

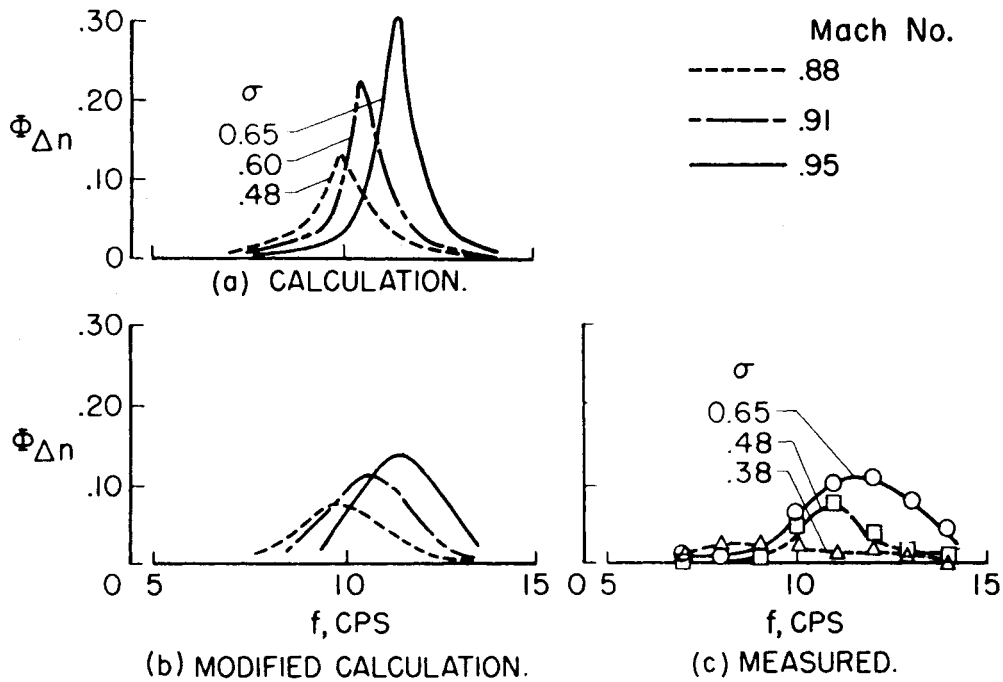


Fig. IVC:13 Comparison of Measured and Calculated Power Spectra of Missile Acceleration in Rough Air

NUMBER OF PEAKS

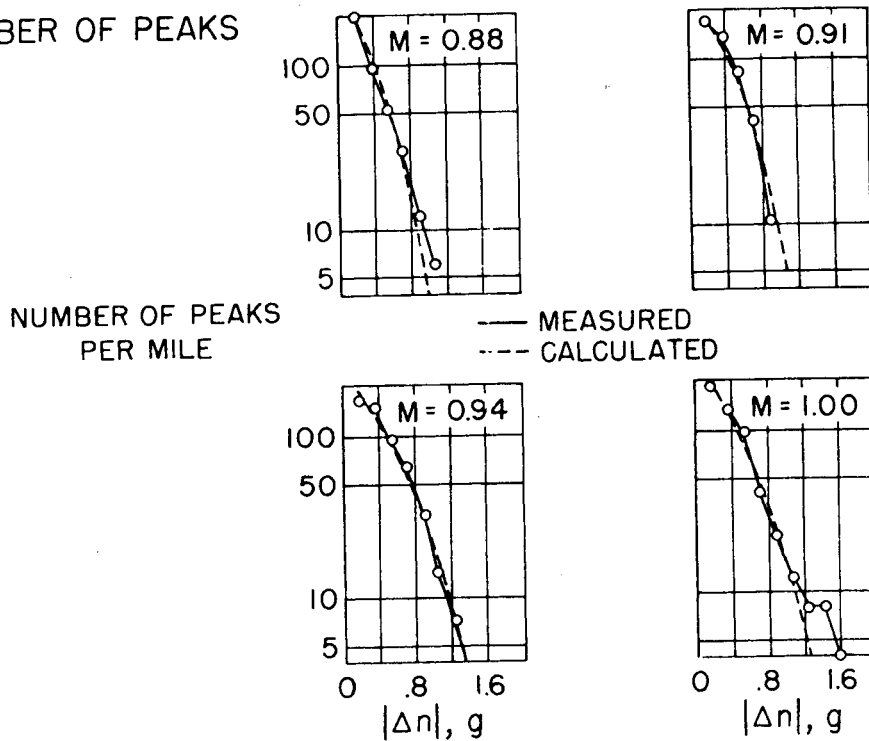


Fig. IVC:14 Comparison of Measured Peak Accelerations in Rough Air with Results Calculated from Measured Power Spectra

calculation outlined in Appendix A. The estimates shown are rather crude in two respects. First, due to the small samples available, roughly two seconds for each spectrum, it was necessary to estimate average power over rather wide frequency band widths. In these cases, each point represents an average over roughly ± 1.5 cps about the values plotted. Second, due to the short record sample available, the statistical reliability for the individual estimates was rather poor. The root-mean-square values are also shown and under the circumstances are considered in reasonable agreement with those calculated.

In order to compare the calculated results with those measured, the calculated spectra were averaged over frequency bands in a roughly equivalent manner to the averaging contained in the spectra measurements and the results obtained are shown on the lower left of Fig. IVC:13. The calculated results with this modification are seen to be in good agreement with those measured and reflect the rather rapid increases in total power with speed, the progressive movement of the spectra to higher frequency, and the increasing peakedness of the spectra. These results thus appear to substantiate the theoretical expectations and have also served to establish the feasibility of the testing and analysis technique in spite of the limitations on sample size.

In order to test the applicability of theoretical results for the Gaussian case for the determination from the power spectrum of the number of peak values exceeding given values, a comparison of the number of accelerations counted from the test records with the number calculated from the measured spectra was made. Some of the results are summarized in Fig. IVC:14. The circled points in each case represent the total number of peak accelerations (maxima and minima) obtained from the records. The dashed curves represent the expected number and were calculated from the measured spectra using Eqs. (33) to (35). The good agreement obtained here and in other such studies has

indicated the Gaussian character of the gust response and has indicated the feasibility of calculating these quantities by spectral techniques.

(c) Effect of Wing Bending Flexibility on Wing Bending Moments in Rough Air

As the next application, Fig. IVC:15 shows some results obtained in a study of the effects of wing bending flexibility on the root bending moment of a B-29 airplane. The figure shows a comparison of the measured and calculated power spectra of root bending moment. In each case, the solid curve is for the flexible airplane and the dashed curve is for the airplane considered as a rigid body. For the rigid case, the measured spectrum was obtained from nodal point acceleration measurements and the conversion to bending moment was based on the calibration data obtained in steady pullups.

In each case, the first peak is associated with the free body mode while the second peak at about 3 cycles per second is associated with the fundamental wing bending mode. The effect of wing bending is in both cases given by the bump in the spectrum at the fundamental wing bending frequency. The ratios of the root-mean-square values are also shown in both cases. Comparison of the measured and calculated results shows rather good agreement in root-mean-square values and in the size and shape of the bump associated with the wing bending effects.

(d) Effects of Variations in Turbulence Across the Airplane Span

Most of the early applications of spectral techniques to turbulence and buffeting problems have assumed the disturbance was one-dimensional and uniform across the span. Recent studies (Refs. 12, 14, and 23) have indicated that the effects of variations in the turbulence across the span may give

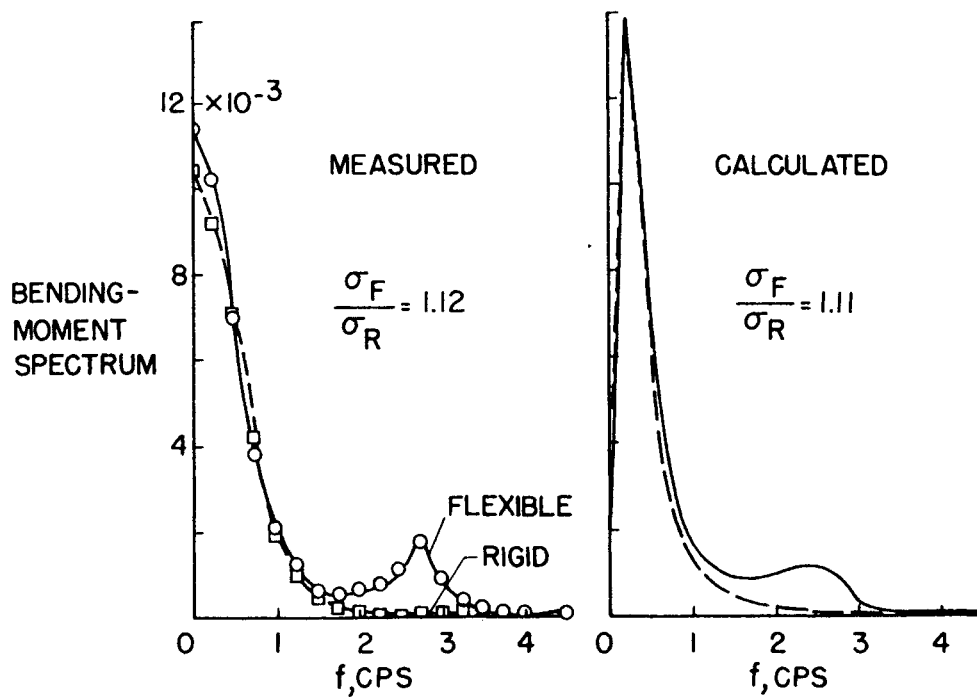


Fig. IVC:15 Effect of Wing Bending Flexibility on Root Bending Moments Due to Gusts

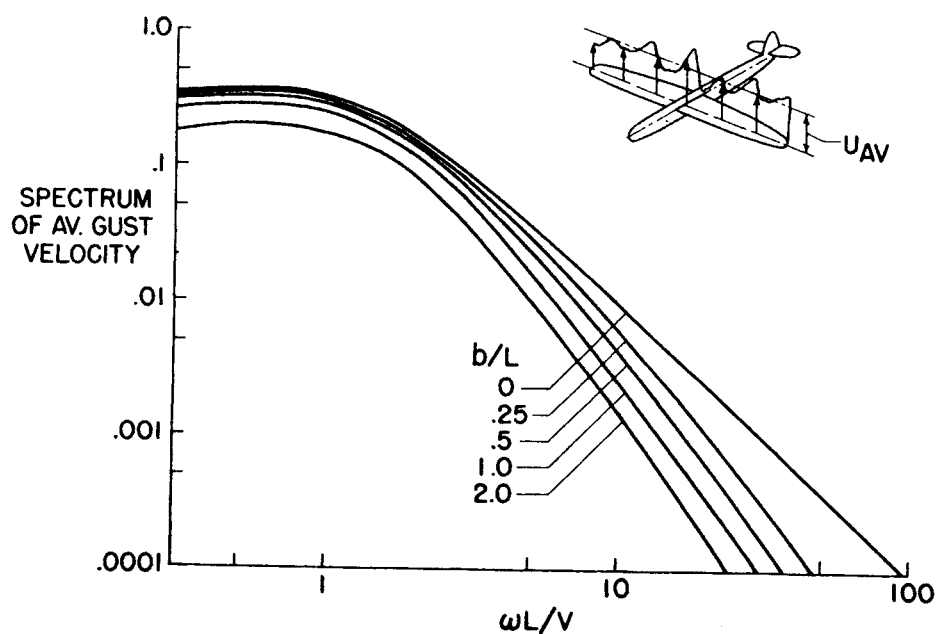


Fig. IVC:16 Effect of Spanwise Variations of Turbulence on the Average Gust Velocity Across the Airplane Span

rise to significant effects in some problems and that the two-dimensional disturbance must be considered in these cases.

Fig. IVC:16, based on the results of Ref. 23, shows the effect of the airplane span length b on the spectrum of the average gust velocity across the span sensed by the airplane. These results are for isotropic turbulence and the analytical expression for the turbulence spectrum given in Fig. IVC:7. The abscissa is the nondimensional frequency $\omega L/V$ where L is the scale of turbulence, and V is the airplane forward speed. Curves are shown for various values of the ratio of the airplane span to turbulence scale b/L ; the curve for $b/L = 0$, giving the spectrum of turbulence at a point, while the other curves give the spectra of average gust velocity across the span. For increasing values of b/L , it can be seen that the effective spectrum is progressively reduced, particularly at the higher frequencies.

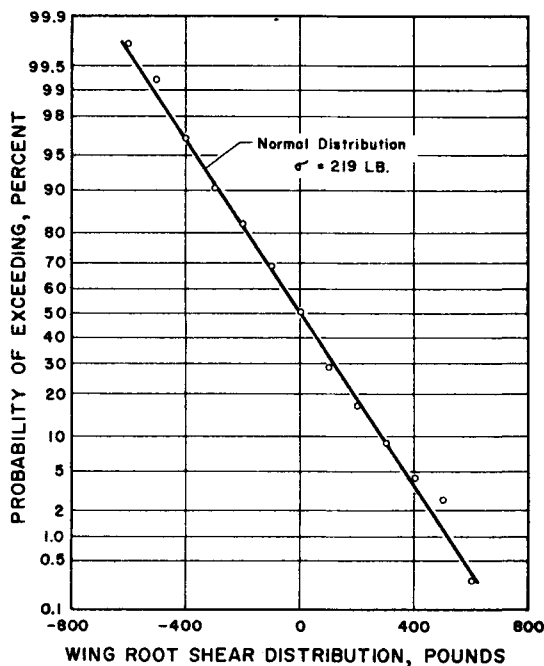


Fig. IVC:17 Probability Distribution of Wing Root Shear Fluctuations in Buffeting Measured During Stall of a Fighter Type Airplane

These results apply directly to wing lift and to overall airplane accelerations and indicate the amount of attenuation due to spanwise gust averaging. The results, while of significance in gust response studies, are of even greater significance in regard to wind tunnel and buffeting studies where the ratio of b/L is likely to be much greater than for the case of atmospheric turbulence.

Although the effects of nonuniform turbulence appear to attenuate the overall airplane load and gust response, they may give rise to a different effect when the internal structural response is considered. Some recent results, Ref. 23, indicate that the bending moments and structural stresses may actually be increased for a two-dimensional disturbance, depending upon the airplane's characteristics.

(e) Buffeting Loads

As a final illustration, some results on buffeting loads obtained in a recent study by W. B. Huston and T. H. Skopinski of NACA are presented. Shown in Fig. IVC:17 is the probability distribution of the wing

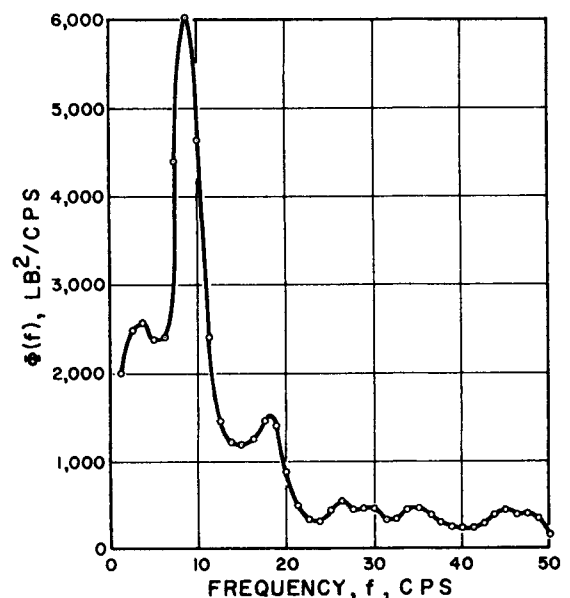


Fig. IVC:18 Measured Power Spectrum of a Wing Shear Fluctuation in Buffeting

shear fluctuations during buffeting as measured by wire resistance strain gages located at the wing root. The measurements, over a period of 3.90 seconds, sampled every 0.01 second, were obtained during a low-speed stall of a fighter airplane.

The fluctuating loads have their origin in the unsteady forces associated with the separated flow over the wing; they represent the response of an elastic system to these random unsteady forces. The probability of exceeding a given value of load is plotted on normal probability paper, a special paper which has the property that the normal probability distribution plots as a straight line. The straight line shown is for a normal distribution with the same standard deviation as the measured distribution (219

pounds) and appears to yield a good fit. These results suggest that the buffet loads may approximate a Gaussian process.

Fig. IVC:18 shows the power spectral density $\Phi(f)$ for the buffet shear loads shown in Fig. IVC:17, as obtained by numerical procedures. The power spectrum is dominated by the response of the wing in the first symmetrical bending mode which, in ground vibration tests, had a frequency of 9.3 cps. Other modes present can be identified such as first asymmetric bending, 18.1 cps, and first torsion, 34.5 cps, but the contribution of these secondary modes is small, and the spectrum is essentially characteristic of the response of a lightly damped single degree of freedom system to a random input.

REFERENCES

1. Clementson, G. C., "An Investigation of the Power Spectral Density of Atmospheric Turbulence," Report No. 6445-T-31, (Sc. D. thesis), Instrumentation Laboratory, Massachusetts Institute of Technology, Cambridge, Massachusetts, May 1950.
2. Liepmann, H. W., "On the Application of Statistical Concepts to the Buffeting Problem," Journal of the Aeronautical Sciences, Vol. 19, No. 12, December 1952.
3. Chippendale, G. R., and Clement, W. F., "A Statistical Study of Atmospheric Turbulence by Flight Measurements," Report No. 6822-2-T-2, (S. M. thesis), Instrumentation Laboratory, Massachusetts Institute of Technology, Cambridge, Massachusetts, May 1951.
4. Fung, Y. C., "Statistical Aspects of Dynamic Loads," Journal of the Aeronautical Sciences, Vol. 20, No. 5, May 1953.
5. Press, H. and Mazelsky, B., "A Study of the Application of Power Spectral Methods of Generalized Harmonic Analysis to Gust Loads on Airplanes," NACA TN 2853, January 1953.
6. Press, H. and Houbolt, J. C., "Some Applications of Generalized Harmonic Analysis to Gust Loads on Airplanes," Paper presented at 22nd Annual Meeting of the Institute of the Aeronautical Sciences, (Preprint No. 449), January 1954.
7. Decaulne, P., "Airplane Lateral Response to Statistical Gust Inputs," Project Meteor Report, (S. M. thesis), Massachusetts Institute of Technology Guided Missiles Program, Cambridge, Massachusetts, May 1952.

8. Vitale, A. J., Press, H., and Shufflebarger, C. C., "An Investigation of the Use of Rocket-Powered Model for Gust-Load Studies with an Application to a Tailless Swept-Wing Model at Transonic Speeds," NACA TN 3161, June 1954.
9. Summers, R. A., "A Statistical Description of Large-Scale Atmospheric Turbulence," Report T-55, (Sc. D. thesis), Instrumentation Laboratory, Massachusetts Institute of Technology, Cambridge, Massachusetts, May 1954.
10. Miles, J. W., "On Structural Fatigue Under Random Loading," Journal of the Aeronautical Sciences, Vol 21, No. 11, November 1954.
11. Chilton, R. G., "Some Measurements of Atmospheric Turbulence Obtained from Flow-Direction Vanes Mounted on an Airplane," NACA TN3313, November 1954.
12. Diederich, F. W., "The Response of an Airplane to Random Atmospheric Disturbances," (Ph. D. thesis), California Institute of Technology, 1954.
13. Mazelsky, B., "Extension of the Power Spectral Methods of Generalized Harmonic Analysis to Determine Non-Gaussian Probability Functions of Random Input Disturbances and Output Responses of Linear Systems," Journal of the Aeronautical Sciences, Vol. 21, No. 3, March 1954.
14. Liepmann, H. W., "Extension of the Statistical Approach to Buffeting and Gust Response of Wings of Finite Span," Douglas Aircraft Company Report SM-15172, February 1954.
15. Houbolt, J. C., "Correlation of Calculation and Flight Studies of the Effect of Wing Flexibility on Structural Response Due to Gusts," NACA TN 3006, August 1953.
16. Walls, J. H., Houbolt, J. C., and Press, H., "Some Measurements and Power Spectra of Runway Roughness," NACA TN 3305, November 1954.
17. Huston, W. B., and Skopinski, T. H., "Measurements and Analysis of Wing and Tail Buffeting Loads and Fighter Type Airplane," NACA TN 3080, 1954.
18. Press, H., Meadows, M. T., and Hadlock, I., "Estimates of Probability Distribution of Root-Mean-Square Gust Velocity of Atmospheric Turbulence by Random-Process Theory," NACA TN 3362, March 1955.
19. Press, H., and Meadows, M. T., "A Reevaluation of Gust Load Statistics for Applications in Spectral Calculations," NACA TN 3540, August 1955.
20. Hess, R. W., Lassiter, L. W., and Hubbard, H. H., "A Study of the Response of Panels to Random Acoustic Excitation," NACA RM L55E13c.
21. North, W. J., "Summary Evaluation of Toothed-Nozzle Attachments as a Jet Noise Suppression Device," NACA TN 3516, July 1955.
22. Notess, C. B., and Eakin, G. J., "Flight Test Investigation of Turbulence Spectra at Low Altitude Using a Direct Method for Measuring Gust Velocities," Cornell Aeronautical Laboratory, Inc., Report No. VC-839-F-1, July 1954.

23. Diederich, F. W., "The Dynamic Response of a Large Airplane to Continuous Random Atmospheric Disturbances," Paper presented at 22nd Annual Meeting of the Institute of the Aeronautical Sciences (Preprint No. 548), January 1955.
24. Connor, R. J., Hawk, J., and Levy, C., "Dynamic Analyses for the C-47 Airplane Gust Load Alleviation System," Report No. SM-14456, Douglas Aircraft Company, Inc., July 29, 1952.
25. Tukey, J., "The Sampling Theory of Power Spectral Estimates," Symposium on Applications of Autocorrelation Analyses to Physical Problems, Woods Hole, Massachusetts, 1949, Office of Naval Research, Washington, D. C.
26. Rice, S. O., "Mathematical Analysis of Noise," Pts. I and II Bell System Technical Journal, Vol. XXIII, No. 3, July 1944, pp. 282-332; Pts. III and IV, Vol. XXIV, No. 1, January 1945, pp. 46-156.
27. Lawson, J. L. and Uhlenbeck, G. E., Editors, "Threshold Signals," McGraw-Hill Book Company, Inc., 1950, Ch. 3.
28. James, H. M., Nichols, N. B., and Phillips, R. S., "Theory of Servo-Mechanisms," McGraw-Hill Book Company, 1947, Chap. 6.
29. Bartlett, M. S., "Introduction to Stochastic Processes," Cambridge University Press, 1955.
30. Blackman, R. B., "Applied Fourier Analysis," Bell Telephone Laboratories.
31. Spetner, L. M., "Errors in Power Spectra Due to Finite Samples," Journal of Applied Physics, May 1954.
32. Smith, F. B., "Analog Equipment for Processing Randomly Fluctuating Data," Paper presented at the Institute of the Aeronautical Sciences annual meeting (Preprint No. 545), January 1955.
33. Kenimer, Robert L., "An Analog Cross-Spectrum Analyzer For Certain Telemetered Data," National Telemetry Conference Record, 1955.
34. Krendel, E. S., Barnes, G. H., "Interim Report on Human Frequency Response Studies," The Franklin Institute, Wright Air Development Center, June 1954.